

A research Report for further analysis to set the

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Executive Summary

This document synthesizes comprehensive research findings across seven topical blocks to establish the scope and research positioning for a Bachelor's thesis comparing 3D Gaussian Splatting (3DGS) with classical Structure-from-Motion/Multi-View Stereo (SfM/MVS) photogrammetry for drone-based 3D reconstruction. The research context encompasses foundational 3DGS theory, the Skyfall-GS satellite imagery pipeline, state-of-the-art UAV photogrammetry, direct comparative studies, georeferencing approaches, physical test field design, and benchmark dataset evaluation protocols.

Key findings establish that 3DGS and classical photogrammetry optimize for fundamentally different objectives: classical SfM/MVS achieves superior metric geometric accuracy (1.43 mm RMSE in controlled conditions), while 3DGS delivers substantially higher visual quality and real-time rendering capability. A critical finding from the GauU-Scene V2 benchmark demonstrates an inverse relationship between image-quality metrics (PSNR) and geometric accuracy (Chamfer distance), establishing that visual rendering quality does not guarantee metric geometric fidelity.

A confirmed research gap exists: no published study has combined a physical Ground Control Point (GCP) testfield measured by total station or RTK GNSS with 3DGS reconstruction and evaluated metric accuracy against surveying-grade ground truth. This gap represents the primary opportunity for original thesis contribution. Additionally, Skyfall-GS—referenced in the initial research brief—is fundamentally a satellite-imagery-only pipeline that has not been evaluated on oblique UAV drone footage, necessitating scope clarification during the thesis proposal phase.

1. Research Scope and Positioning

The proposed thesis contributes to the convergence of computer vision (3D reconstruction via neural representations) and geodetic measurement (metric accuracy validation via surveying control networks). This positioning is distinct from the majority of existing 3DGS literature, which prioritizes rendering quality without explicit geodetic validation.

The research addresses evaluation dimensions across six criteria:

- **D1 - Geometric Accuracy:** Metric-scale deviation from surveying ground truth (RMSE, M3C2 distance, Chamfer distance)
- **D2 - Visual Quality:** Rendering metrics (PSNR, SSIM, LPIPS) and appearance fidelity
- **D3 - Manual Object Identifiability:** Human-assessed visibility and recognizability of objects in rendered views
- **D4 - ML Object Detection Performance:** Automated object recognition metrics (mAP) across reconstruction outputs
- **D5 - Practical Applicability:** Processing speed, hardware requirements, costs, licensing, and GIS integration

This multidimensional framework directly addresses the research brief's core requirement for a "systematic comparison via Skyfall-GS Pipeline" focused on "speed, metric accuracy, and costs" with "evaluation of geometric accuracy, visual quality, object detectability."

2. 3D Gaussian Splatting: Theoretical Foundations

2.1 Foundational Architecture and Paradigm

Kerbl, Kopanas, Leimkühler, and Drettakis (2023) introduced "3D Gaussian Splatting for Real-Time Radiance Field Rendering" in ACM Transactions on Graphics, Volume 42(4), awarded Best Paper at SIGGRAPH 2023. The method departs fundamentally from neural radiance field (NeRF) approaches by employing explicit geometric primitives—anisotropic 3D Gaussian ellipsoids—initialized from Structure-from-Motion sparse point clouds, rather than continuous volumetric representations encoded in multilayer perceptrons.

Three core innovations define the architecture:

- 1. Anisotropic 3D Gaussian Scene Representation:** Scenes are represented as collections of 3D Gaussians that preserve volumetric radiance field properties while avoiding computation in empty space, enabling computational efficiency relative to voxel-based approaches.
- 2. Interleaved Optimization with Adaptive Density Control:** The training process alternates between optimizing Gaussian positions, covariances, opacities, and spherical harmonic coefficients, with adaptive densification phases that clone Gaussians in underrepresented regions and prune low-contribution primitives.
- 3. Tile-Based Differentiable Rasterizer:** A GPU-native rendering algorithm implements anisotropic splatting through rasterization rather than ray-marching, achieving real-time performance exceeding 100 frames-per-second (FPS) on consumer hardware at 1080p resolution, a dramatic performance advantage over NeRF's typical 5–10 FPS.

The fundamental geodetic limitation inherent in this architecture must be emphasized: 3DGS optimizes photometric loss—the L1 or SSIM difference between rendered and observed pixel values—rather than geometric conformity to physical surfaces. Gaussian primitives can deform, overlap, and aggregate in configurations that appear visually correct to human observers while deviating metrically from underlying geometry, particularly in textureless regions, under vegetation occlusion, and at object boundaries.

2.2 Variant Architecture Landscape

Five variant architectures are directly relevant to thesis evaluation:

Mip-Splatting (CVPR 2024, Best Student Paper): Addresses aliasing artifacts when camera distance or focal length changes—critical for UAV imagery spanning multiple altitudes. Two components implement this: a 3D Smoothing Filter constraining Gaussian primitive sizes based on maximal sampling frequency from input views, and a 2D Mip Filter approximating physical imaging box filters. Performance: average PSNR 34.56 dB versus vanilla 3DGS 28.79 dB on multi-scale Blender dataset.

2D Gaussian Splatting (SIGGRAPH 2024): Collapses 3D volume into 2D oriented planar Gaussian disks, providing view-consistent geometry via ray-splat intersection and rasterization. Geodetically significant finding: on transparent objects, 2DGS achieved RMSE 0.00121 m versus 3DGS+SuGaR's 0.00125 m—superior geometric accuracy despite inferior rendering metrics (PSNR 34.91 dB vs. 36.06 dB), with lower standard deviation indicating more spatially consistent accuracy.

Scaffold-GS (CVPR 2024, Highlight): Uses anchor points from sparse voxel grids with neural Gaussians spawned via learnable offsets, achieving 4–10× storage reduction (156 MB vs. 693 MB on Mip-NeRF360, 87 MB vs. 411 MB on Tanks&Temples) while maintaining ~100 FPS rendering and

comparable PSNR (28.84 vs. 28.69 on Mip-NeRF360).

SuGaR (CVPR 2024): Introduces surface-alignment regularization encouraging Gaussians to align with scene surfaces, followed by Poisson reconstruction for clean mesh extraction in 5–10 minutes per scene. Geodetically relevant as an enabler of mesh output for CAD/GIS integration, though mesh extraction introduces additional approximation errors.

GaussianPro (ICML 2024): Addresses 3DGS's dependence on SfM initialization in textureless regions through progressive propagation using MVS patch-matching. On the Waymo large-scale outdoor dataset (scene 1002751): 35.97 PSNR, 0.959 SSIM, 0.207 LPIPS versus vanilla 3DGS's 35.22 PSNR—a 1.15 dB improvement, though training increases from 40 to 56 minutes.

The open-source **gsplat library** (PyTorch+CUDA, available on GitHub) provides 10% less training time and 4× less memory than the original reference implementation, with integrated support for multiple densification strategies and Mip-Splatting anti-aliasing—making it the recommended implementation for thesis work with typical GPU hardware.

3. Critical Scoping Issue: Skyfall-GS as Satellite-Only Pipeline

3.1 Full Citation and Architecture Assessment

Lee, J.-Y., Liu, Y.-R., Tsai, S.-R., and colleagues (2025) published "Skyfall-GS: Synthesizing Immersive 3D Urban Scenes from Satellite Imagery" as an arXiv preprint (2510.15869) on October 17, 2025. The work is currently under conference review (OpenReview) and has not yet undergone formal peer review. The GitHub repository (github.com/jayin92/skyfall-gs) has accumulated 843 stars, indicating community interest.

Critical Finding: Skyfall-GS is fundamentally architected for near-nadir satellite imagery with extremely limited parallax (WorldView-3, GoogleEarth). The paper contains no evaluation on oblique UAV or drone footage of any kind.

The two-stage pipeline consists of:

Stage 1 (Reconstruction, approximately 1 hour on NVIDIA RTX A6000): Enhanced 3DGS with three satellite-specific components:

- Pseudo-camera depth supervision using MoGe monocular depth estimation from virtual cameras at reduced elevations (80° down to 45°), sampled every 10 iterations across 24 views
- WildGaussians-adapted appearance modeling MLP with per-image and per-Gaussian embeddings for handling multi-date illumination variation
- Entropy-based opacity regularization preventing floating artifacts

Stage 2 (Curriculum refinement, approximately 6 hours on RTX A6000): Five episodes of iterative diffusion-based refinement using FLUX.1 [dev] text-to-image model with FlowEdit prompt-to-prompt editing, progressively descending camera elevation from 85° to 45° to synthesize realistic ground-level details.

3.2 Performance and Limitations

On DFC2019 (Jacksonville, Florida; WorldView-3 satellite imagery), Skyfall-GS achieved FIDCLIP 27.35 and CMMD 2.086, substantially outperforming baselines (Sat-NeRF 88.36, EOGS 87.74). On GoogleEarth (New York City), FIDCLIP 9.91 and CMMD 2.009. Rendering achieved 11 FPS on NVIDIA T4 and 40 FPS on MacBook Air M2. Human evaluation with 89 participants yielded approximately 97% win rate on DFC2019 and 90% on GoogleEarth.

However, critical limitations directly preclude direct application to UAV workflows:

1. **No UAV/Drone Evaluation:** The pipeline has never been tested on oblique aerial imagery or drone video capture. The pseudo-camera depth supervision component is specifically designed for limited-parallax nadir satellite views and may not transfer to high-parallax drone footage without substantial architectural modification.
2. **Absence of Geodetic Validation:** Skyfall-GS uses distribution-based metrics (FIDCLIP, CMMD) and subjective user studies. No evaluation against Ground Control Points, no RMSE computation against surveying instruments, no geometric accuracy assessment exists in the paper.
3. **Generative Hallucination Risk:** The FLUX.1 [dev] diffusion refinement stage synthesizes details that may be inconsistent with actual ground-level geometry, particularly in regions with limited satellite coverage. Community feedback confirms: "when you bring the viewpoint down to near street level and look horizontally, it's worse than traditional photogrammetry methods."
4. **Over-Smoothed Street-Level Textures:** Acknowledged limitation in the paper for near-horizontal viewing angles.

3.3 Thesis Scope Clarification Required

The initial research brief specified Skyfall-GS as the focal comparison pipeline for "oblique aerial and satellite images." The published paper addresses exclusively nadir satellite data. This represents a fundamental scoping discrepancy that must be resolved during the thesis proposal phase through one of two approaches:

Option A (Recommended for timeline): Reframe the comparison to use drone-specific 3DGS pipelines (DroneSplat for CVPR 2025, standard INRIA 3DGS implementation, or DJI Terra V5.0 for commercial baseline) with Skyfall-GS mentioned as satellite-specific context and architectural reference.

Option B (Higher contribution potential): Design an original experiment adapting Skyfall-GS to oblique drone data with explicit notation that this constitutes novel application research beyond the published paper's scope, including architectural modifications and validation against UAV-specific benchmarks.

Option A reduces technical risk for a bachelor's thesis timeline while maintaining rigorous comparative methodology. Option B offers stronger original contribution but requires careful scoping to remain feasible within one-year thesis duration.

4. Classical UAV Photogrammetry: Accuracy Benchmarks and Software State of the Art

4.1 Software Accuracy and Performance Comparison

A 2024 processing benchmark using an identical 420-image dataset (DJI Mavic 3 Enterprise, 75% front overlap, 65% side overlap, 80 meter altitude) provides the most direct software performance comparison:

Software Performance Benchmark (2024)

Software	Checkpoint RMSE	Point Cloud Density	Processing Time	License Cost (2026)
Agisoft Metashape Pro	1.9 cm	389 pts/m ²	2h 12m alignment + processing	\$3,499 perpetual
Pix4D Mapper	1.8 cm	412 pts/m ²	2h 47m total	\$4,200/year (cloud)
DroneDeploy	2.4 cm	298 pts/m ²	54 minutes (cloud processing)	\$4,000–\$12,000/year
OpenDroneMap (WebODM)	2.7 cm	341 pts/m ²	3h 38m	Free (self-hosted)
RealityCapture	—	—	10–20× faster than Metashape (datasets >1,000 images)	€250/3 months (PPI model)

Metashape Pro achieves financial break-even with Pix4D's subscription model within 10 months of perpetual licensing. DroneDeploy's cloud-first architecture eliminates workstation hardware investment and labor monitoring overhead, achieving approximately 50% lower cost-per-deliverable (\$36.93 versus \$72.77 for desktop alternatives) when processing 15 or more projects monthly.

COLMAP, the open-source reference implementation for maximum precision, requires specialized workflow knowledge but provides centimeter-level accuracy at zero licensing cost, making it the standard for research environments.

4.2 GCP-Based Georeferencing Accuracy Relationships

The foundational GCP study (MDPI Remote Sensing 2018, analyzing 3,465 different GCP combinations across 1,225 hectares) establishes critical relationships between GCP count, distribution, and achievable accuracy:

GCP Density vs. RMSE Performance:

- 10–20 GCPs: RMSE $\approx \pm 31$ cm (approximately 5× Ground Sampling Distance)
- 50–60 GCPs (optimal range): RMSE $\approx \pm 16$ cm (approximately 3× GSD)
- 90–100 GCPs: RMSE $\approx \pm 12$ cm (approximately 2× GSD)
- 3–5 GCPs per 100 photographs: Planimetric RMSE $\approx \pm 1\times$ GSD

Vertical errors are systematically approximately 2.5 times greater than horizontal (easting/northing) errors. This asymmetry reflects the geometry of aerial photography, where vertical error is more sensitive to camera calibration uncertainty and baseline geometry.

Critical Methodological Distinction: RMSE measured only on control points (GCPs used in bundle adjustment) overestimates true accuracy by factors of 3 to 6 times compared to independent check points that are explicitly excluded from processing. With as few as 3–5 GCPs, control point residuals may show millimeter-level accuracy while independent check points reveal actual RMSE of ± 1 –8 meters.

4.3 RTK and Direct Georeferencing

Real-Time Kinematic (RTK) GNSS-equipped UAVs provide direct georeferencing without physical GCPs:

- DJI Phantom 4 RTK (2019–2020 studies): 1–3 cm horizontal accuracy, 4–6 cm vertical accuracy
- Santa Rita Experimental Range deployment (2019): 2.2 cm horizontal RMSE, 3.4 cm vertical RMSE
- RTK + oblique imagery (10–12% of nadir images) without GCPs: 1.26 cm XY RMSE, 5.58 cm Z RMSE

Camera angle emerges as more critical than flying altitude for spatial error structure optimization. Higher camera inclinations (20–40°) reduce both error magnitude and spatial autocorrelation (Moran's I statistic), suggesting that oblique flight planning is an essential component of geodetic-quality UAV workflows.

4.4 Video-Based Structure from Motion

Video-based photogrammetry reduces field capture time by 50–70% compared to still image acquisition. The 2024 Wuhan University VSLAM+SfM hybrid (DJI M300 RTK with DJI P1, 3840×2160 resolution at 60 FPS) achieves 100–200% efficiency improvement versus COLMAP and 50–100% efficiency improvement versus Theia, while maintaining centimeter-level accuracy under GCP constraints. For consumer-class drones like the DJI Mini 2, image quality between raw video screenshots and dedicated still photography is negligible from photogrammetric perspective.

5. Direct Comparative Studies: 3DGS versus Classical Photogrammetry

5.1 ISPRS Annals 2025: Most Geodetically Relevant Peer-Reviewed Comparison

Petrovska and Jutzi (Karlsruhe Institute of Technology, published in ISPRS Annals of Photogrammetry, Remote Sensing and Spatial Information Sciences, April 2025) conducted the single most geodetically meaningful published comparison, evaluating four 3DGS implementations against classical MVS and NeRF-based methods on the STELLA dataset. The study specifically addresses vegetation occlusion as a critical real-world scenario for forestry and agricultural monitoring applications.

Geometric Accuracy Comparison (RMSE in Millimeters)

Scenario 1: No Occlusion

- MVS: **1.43 mm** (best overall accuracy)
- 3DGS-Basic: 4.74 mm
- NeRF: 5.36 mm
- 3DGS-MCMC: 5.80 mm
- 3DGS-LumaAI: 7.02 mm
- Splatfacto: 7.28 mm (worst among methods)

Scenario 2: Vegetation Occlusion

- MVS: **3.23 mm** (maintains superiority)
- 3DGS-LumaAI: 7.58 mm (best 3DGS method under occlusion)
- 3DGS-Basic: 7.49 mm
- NeRF: 12.67 mm
- Splatfacto: 13.80 mm
- 3DGS-MCMC: 17.36 mm (significant degradation under occlusion)

Completeness Assessment

Without Occlusion:

- 3DGS-MCMC: 99.45% (excellent volumetric completeness)
- NeRF: 98.23%
- MVS: 97.35%
- 3DGS-Basic: 82.75%
- Splatfacto: 79.83%

With Vegetation Occlusion:

- MVS: 75.95% (classical method advantage under real-world conditions)
- NeRF: 69.95%
- 3DGS-MCMC: 65.93% (severe degradation: 33.5 percentage point loss)
- 3DGS-Basic: 42.71%
- Splatfacto: 31.80% (worst performance)

Point cloud density results show 3DGS-MCMC reconstructing 1,556,040 total points versus 845,456 for MVS and 352,298 for 3DGS-Basic, suggesting that higher Gaussian count does not necessarily translate to better geometric accuracy or robust handling of occlusion.

5.2 GauU-Scene V2 Benchmark: Paradigm-Shifting Inverse Relationship

The GauU-Scene V2 benchmark (University of Science and Technology of China, arXiv:2404.04880) represents the largest outdoor LiDAR-image dataset for UAV 3DGS evaluation (6 scenes, 6.668 km² total, 4,693 images, 627.5 million LiDAR points). This benchmark reveals a critical finding that should anchor any comparative thesis framework:

The inverse relationship between image-quality metrics (PSNR, SSIM, LPIPS) and geometric accuracy (Chamfer distance):

Method	PSNR (dB)	SSIM	LPIPS	Chamfer Distance (m)	GPU Training Time
Vanilla 3DGS	24.42 (best)	0.752 (best)	0.300 (best)	0.133 m	63.7 min
NeRFacto	17.04 (worst)	0.422 (worst)	0.764 (worst)	0.093 m (best geometry)	1,735.3 min
InstantNGP	19.74	0.510	0.815	0.149 m	—
SuGaR	—	—	—	0.189 m (worst)	—

This finding is fundamentally important: 3DGS optimizes photometric loss (appearance matching), not geometric surface conformity. NeRFacto, while producing visually inferior renders (PSNR 17.04 dB versus 3DGS's 24.42 dB), achieves superior metric accuracy (0.093 m versus 0.133 m Chamfer distance) because its implicit volumetric representation is constrained differently. The inverse correlation demonstrates that high rendering quality does not guarantee geometric fidelity—a critical insight for geodetically motivated thesis work.

5.3 UAV M3C2 Geometric Analysis

Comparative research on UAV imagery (MDPI Sensors 2025, PMC12115230) using Multiscale Model-to-Model Cloud Comparison (M3C2) analysis provides ground-truth SfM-MVS validation:

- Study Area 1 photogrammetric accuracy: RMSE_X 0.026 m, RMSE_Y 0.035 m, RMSE_Z 0.141 m
- Study Area 2: RMSE_X 0.020 m, RMSE_Y 0.017 m, RMSE_Z 0.240 m

Splatfacto (3DGS via Nerfstudio) rendering quality on Study Area 2: PSNR 28.93 dB, SSIM 0.9180, LPIPS 0.0561, substantially outperforming Nerfacto (PSNR 20.57 dB, SSIM 0.4954) and Instant-NGP (PSNR 24.80 dB, SSIM 0.7715). M3C2 geometric error shows Splatfacto achieving approximately 0.2 meter average error versus sub-decimeter accuracy from classical MVS reference, with NeRF point clouds demonstrating increased noise as image resolution decreases.

5.4 Transparent Object Reconstruction: Visual vs. Geometric Trade-off

MDPI Sensors 2025 comparative study on transparent glass objects provides the clearest demonstration of the visual/geometric quality disconnect:

Method	Cloud-to-Mesh RMSE	Standard Deviation	Completeness	PSNR (dB)
2D Gaussian Splatting	0.00121 m (best geometry)	0.0005 m (most consistent)	96.43%	34.91
3DGS + SuGaR	0.00125 m	0.0011 m	99.62% (best completeness)	36.06 (best rendering)
Agisoft Metashape	0.00161 m	0.0008 m	16.96% (worst)	—

2DGS demonstrates superior geometric accuracy (RMSE 0.00121 m vs. 0.00125 m) and spatial consistency (lower standard deviation) despite inferior rendering quality (PSNR 34.91 dB vs. 36.06 dB). This inversion of expected relationship—better geometry from the method with worse rendering metrics—is methodologically significant and supports thesis argument that rendering and geometric quality must be evaluated as distinct dimensions.

6. 3DGS on UAV-Specific Aerial Data

6.1 Drone-Specific 3DGS Implementations

Rapid development since 2024 has produced multiple UAV-specialized 3DGS variants addressing core challenges of unbounded scenes, dynamic objects, limited parallax, and large-scale partitioning.

DroneSplat (CVPR 2025): Tang et al. directly address in-the-wild drone reconstruction through Adaptive Local-Global Masking for dynamic object elimination and Multi-View Stereo predictions with voxel-guided optimization. The associated dataset includes 24 sequences captured with DJI Mavic Pro 3 at 1920×1080 and 3840×2160 resolutions. Performance on dynamic scenes: "Low Dynamic" Cultural Center (PSNR 24.53 SSIM 0.785 LPIPS 0.149 vs. prior best GS-W at 23.29 PSNR); "High Dynamic" Simingshan (PSNR 24.17 SSIM 0.808 LPIPS 0.093 vs. vanilla 3DGS at 22.14 PSNR). Under six-view limited input conditions, DroneSplat achieves PSNR 18.09–18.17 with SSIM 0.496–0.534, outperforming Scaffold-GS, FSGS, and InstantSplat.

HUG: Hierarchical Urban Gaussian Splatting (ICCV 2025): Implements octree-based Level-of-Detail with visibility-based partitioning (threshold 800 visible points per block), reducing preprocessing from approximately 2 hours to approximately 1 minute per scene on MatrixCity dataset. Achieves PSNR 31.12 versus CityGS baseline 27.23; SSIM 0.912 versus 0.806; LPIPS 0.106 versus 0.307.

AGS: Aerial Gaussian Splatting (2024): First application of 3DGS to large-scale aerial MVS, evaluated on WHU-OMVS dataset (10-cm GSD, 550 m altitude, 268 images, 850×700 m² coverage) and Tianjin dataset (20-cm GSD, 200 m altitude, 342 images, 400×350 m² coverage). AGS surpassed GOF by 1.5% and COLMAP by 9.4% in highest-resolution PAG metrics.

DJI Terra V5.0 (July 2025): First consumer-grade 3DGS integration by major drone manufacturer. Processing capacity: approximately 500 images/hour versus 200 in Terra 4.0; 1,500-image projects complete in <3 hours versus 6–8 hours with mesh technology. Memory efficiency: 300–400 images

per gigabyte RAM; 64 GB workstation handles 20,000–25,000 images; 128 GB handles 40,000 images. Optimized for RTK-equipped platforms (Matrice 4E) and high-resolution payloads (Zenmuse P1 45 MP, Zenmuse L2 LiDAR).

ULSR-GS: Handles scenes exceeding 1 km² through point-to-photo partitioning and multi-view geometric consistency constraints with adaptive densification. Evaluated on GauU-Scene (>1 km² each, 1,200 oblique images per region), UrbanScene3D (5,000 Penta-Cam images within 2.5 km), and custom datasets (4 km² campus, 8,000+ images; 2 km² complex urban, 10,000+ oblique images).

6.2 GauU-Scene V2: Critical Benchmark for UAV 3DGS

The GauU-Scene V2 benchmark provides the most relevant large-scale UAV 3DGS dataset with LiDAR ground truth. Captured with DJI Matrix 300 + Zenmuse L1 LiDAR sensor across six scenes totaling 6.668 km²:

- Lower Campus: 1.020 km², 670 images, 79.8 million points
- Upper Campus: 0.923 km², 715 images, 94.2 million points
- HAV: 0.815 km², 424 images, 26.8 million points
- LFLS: 1.467 km², 1,106 images, 98.5 million points
- SMBU: 0.908 km², 563 images, 283.3 million points
- SZIT: 1.557 km², 1,215 images, 59.0 million points
- **Total:** 4,693 images, 627.5 million LiDAR points, 92.1 GB raw data

This dataset represents the current state-of-the-art large-scale UAV 3DGS evaluation resource and is recommended for external validation of thesis results.

7. Georeferencing 3DGS Models: Solutions and Standards

7.1 GeoRefGS: Academic Solution with Simulation-Based Validation

Hou, Zhang, Li, and Cui (Wuhan Early Warning Academy, 2026) published "GeoRefGS: Towards Georeferenced 3D Gaussian Splatting from Unmanned Aerial Vehicle Platforms" in Drones, Vol. 10, Issue 3, Article 195, March 11, 2026 (DOI: 10.3390/drones10030195). This is the first peer-reviewed academic method integrating GCP constraints directly into 3DGS training as intrinsic constraints rather than post-processing.

Methodology employs: (1) learnable similarity transformation matrix T connecting local model coordinates to global geographic coordinates; (2) multi-tiered geographic loss functions combining absolute positioning loss, GCP constraint loss, and relative geometric constraints; and (3) adaptive weighting mechanism preventing geographic gradients from overwhelming early optimization iterations.

Accuracy results:

- Mean planar error: 0.054 m
- Mean vertical error: 0.053 m
- PSNR improvement: +3.31 dB over baseline 3DGS
- Improvement over SfM-BBA baseline: 83% reduction (0.32 m → 0.054 m)

Ablation study demonstrates GCP constraint loss alone reduces planar error from 25.1 m to 0.0145 m, establishing GCPs as essential for sub-decimeter 3DGS georeferencing accuracy.

Critical Limitation: Validation conducted in simulated Unity environment, not real drone captures. This represents the primary research gap the proposed thesis directly addresses.

7.2 Commercial Implementation

Pix4D (PIX4Dcatch): October 2025 announcement of georeferenced 3DGS integration supporting 50–2,000 image datasets with or without RTK. Maintains visibility of thin structures (scaffolding, reinforcement mesh) that disappear in classical mesh noise.

Cesium Ion: End-to-end automatic reconstruction as Mesh, PointCloud, or 3DGS with demonstrated capability on large datasets (Microsoft Redmond Campus: 20,169 photos covering 3.7 km² with 110 million splats at 3 cm GSD).

Open-Source Pipeline: GitHub implementation (manudelu/georeferenced_gsplat) extracts GPS metadata from drone images, aligns via COLMAP SfM to East-North-Up coordinates using first image as reference, with Cesium Ion/CloudCompare/Unreal Engine compatibility.

7.3 Standardization Status (2026)

Multiple standardization initiatives define the 2026 landscape:

glTF KHR_gaussian_splatting: Released as specification (February 2026 Release Candidate, ratification expected Q2 2026) by Khronos Group, OGC, Niantic Spatial, Cesium, and Esri. Stores 3D Gaussian data including XYZ position, quaternion rotation (encoded as 10-bit signed integers), anisotropic ellipsoid scale, opacity, and spherical harmonics coefficients.

SPZ Compression Format: Niantic Spatial MIT-licensed compression achieving up to 90% file size reduction versus uncompressed PLY while maintaining visual quality.

OGC 3D Tiles 2.0: Incorporates 3DGS within geospatial standardized tiling framework; Cesium uses 3D Tiles as spatial index with glTF payload for streaming large 3DGS datasets with Level-of-Detail management.

OpenUSD Integration: Alliance for OpenUSD (Pixar, Apple, NVIDIA, Adobe) v26.03 includes UsdVolParticleField3DGaussianSplat schema for native 3DGS representation.

7.4 GIS Platform Integration (2026)

Integration status across GIS platforms remains incomplete:

- **QGIS 3.34+:** Production support for 3D Tiles via QgsTiledSceneLayer
- **QGIS 4.4 (planned):** Native 3DGS extension in development, sponsored by project
- **ArcGIS Pro:** 3D Tiles and I3S support via CesiumJS integration; constraint to WGS84 coordinate systems
- **CesiumJS:** Full LOD streaming for 3DGS (April 2026 update)
- **CloudCompare:** Indirect support via PLY import for M3C2 geometric analysis
- **CityGML:** No documented integration
- **Potree:** No documented integration

These integration gaps represent opportunities for geodetic workflow enhancement in thesis work.

8. Physical Test Field Design and Measurement Standards

8.1 Ground Control Point Specification

GCP markers must employ high-contrast patterns (checkerboard or L-shaped, not X-shaped which introduces up to 10 cm measurement error) with matte surfaces avoiding sun glare and reflections. Material selection should prioritize durability and weather resistance (vinyl floor tiles, stenciled concrete, UV-resistant plastic).

Critical sizing relationship: Target dimensions must be **5 to 10 times the Ground Sampling Distance**. Practical examples:

- At 2 cm/pixel GSD: targets 10–20 cm wide
- At 5 cm/pixel GSD: targets minimum 25–50 cm
- Absolute minimum for typical UAV operations: 0.5 m × 0.5 m
- For permanent painted markings: minimum 600 mm per dimension

8.2 Optimal GCP Density and Distribution

GCP requirements scale with survey area extent:

Survey Area Size	Minimum GCPs	Recommended	Density
<5 hectares (12 acres)	3	5–8	Baseline triangulation
5–20 hectares	5	7–10	Standard projects
20–80 hectares	8	12–15	Larger coverage
>80 hectares	10	15–20+	Complex geometry

Optimal density: 0.5–1.8 GCPs per hectare depending on terrain complexity. Empirical finding from corridor mapping study (2.1 km × 190 m road): 9–11 GCPs (4.3–5.2 GCPs/km) in alternating offset pattern on both sides plus endpoint pairs achieves RMSE_XY 0.028–0.029 m and RMSE_Z 0.055–0.057 m. Significantly diminishing returns appear beyond 5–10 GCPs, with additional points showing logarithmic accuracy improvement.

Distribution strategy ("tent stake" concept): Place GCPs first at survey area perimeter, then distribute across interior. For significant terrain relief, position GCPs at highest and lowest elevation points. Minimum spacing: 100 meters between GCPs.

8.3 Measurement Accuracy Standards

RTK (Real-Time Kinematic) GNSS Requirements:

- Horizontal and vertical accuracy: ≤0.02 m (2 cm)
- Solution status: Fix (not Float)
- Estimated error: <3 cm (HRMS)
- Data collection: Two separate sessions of 30 measurements at 1-second intervals per point
- Typical RTK systems (Leica GS-Series, Topcon HiPer-Series, Emlid Reach RS2): achieve 1–3 cm horizontal accuracy in normal conditions

Post-Processing Kinematic (PPK) Alternative: Equivalent centimeter-level accuracy as RTK, applied post-fieldwork using base station corrections; preferred in areas with poor cellular coverage for real-time RTK corrections.

Total Station: Millimeter-level angular precision for sub-centimeter GCP accuracy requirements or GPS-challenged locations.

Critical Operational Details:

- Antenna height must be precisely measured and entered into rover
- Surveying rod must remain absolutely vertical
- Level bubble must remain centered during data capture
- Rover pole tip must center precisely on target crosshair
- GCP center must localize to pixel-level precision in drone images
- Minimum 5–6 image overlaps recommended per GCP
- Maintain minimum 150-pixel distance from image frame boundaries

8.4 Quality Control Points vs. Ground Control Points

Methodologically critical distinction: **Quality Control Points (QCPs) must be deliberately excluded from processing** and serve as independent validation. Without independent check points, reported accuracy overestimates true performance by factors of 3–6 times. Standard practice: designate portion of physical markers as control points and remainder as check points.

8.5 RTK/PPK Integration with GCP Requirements

RTK/PPK-equipped drones substantially reduce GCP requirements:

- RTK-only requirement: only 4–5 GCPs per square mile (2.6 km²) for elevation variation
- High-contrast urban environments: potentially only 3 well-distributed control points
- Hybrid approach (RTK/PPK + few GCPs): provides both speed of direct georeferencing with accountability of traditional control networks

9. Benchmark Datasets for External Validation

9.1 Standard Computer Vision Benchmarks

Five primary benchmark datasets define the current evaluation landscape:

Tanks and Temples (SIGGRAPH 2017): 14 scenes with FARO industrial laser scanning ground truth (0.1 mm uncertainty at 10.2 m). F-score metric combines precision and recall. Available at tanksandtemples.org. Object-to-building scale evaluation; not UAV-specific but widely referenced for 3DGS validation.

ETH3D (CVPR 2017): 13 training + 12 test scenes with laser scan ground truth across 24 MP DSLR imagery. Voxel-based evaluation protocol prevents artificial inflation of completeness metrics. Distance thresholds from 1 cm to 50 cm. Available at eth3d.net. Indoor and outdoor diversity; good for generalization testing.

DTU (124 scenes): Structured light scanning with STL reference point clouds. 80 scenes used in standard evaluation. Image resolution 1600×1200. Training includes 27,097 total images. Good for small-object accuracy assessment but limited to laboratory-controlled conditions.

MipNeRF360 (9 scenes): 5 outdoor + 4 indoor scenes with no geometric ground truth—rendering-only benchmark. PSNR/SSIM/LPIPS metrics exclusively. **Critical gap:** cannot assess geometric accuracy against this dataset.

BlendedMVS (CVPR 2020): 113 scenes with 17,818 total high-resolution images. Synthetic dataset generated through rendering photorealistic scenes, then blending with input images. Excellent for generalization testing across diverse urban and architectural scenarios.

9.2 UAV-Specific and Geodetically Relevant Benchmark: GauU-Scene V2

GauU-Scene V2 is strongly recommended for thesis external validation as it provides both rendering and geometric metrics for UAV-captured imagery with LiDAR ground truth. The dataset's key characteristics:

- **Coverage:** 6.668 km² across 6 distinct geographic regions
- **Imagery:** 4,693 photographs total, capture equipment: DJI Matrix 300 + Zenmuse L1 LiDAR
- **Ground Truth:** 627.5 million LiDAR points providing dense geometric reference
- **Metrics Available:** PSNR/SSIM/LPIPS (rendering), Chamfer distance (geometry), training efficiency
- **Size Span:** Extends from building-scale to large urban campus scale

Internal validation findings: NeRFacto achieved best geometric accuracy (0.093 m Chamfer distance) while vanilla 3DGS achieved best rendering (PSNR 24.42 dB)—directly supporting thesis argument that rendering and geometric quality operate as distinct optimization objectives.

9.3 NERFBK: Multi-Category Evaluation Framework

NERFBK (ISPRS 2023) provides multi-category comparison across industrial, transparent (refractive), heritage, and aerial dataset categories with different surface characteristics (featureless, well-textured, refractive, reflective). Less mature than Tanks & Temples but broader in scope, including aerial imagery evaluation.

10. Confirmed Research Gaps and Thesis Positioning

10.1 Primary Research Gap: Physical GCP Testfield Validation of 3DGS

No published study has combined a physical Ground Control Point testfield measured by total station or RTK GNSS with 3DGS reconstruction and evaluated metric accuracy against surveying-grade ground truth.

The closest published work (GeoRefGS, 2026) validated accuracy exclusively in a Unity simulated environment without real drone video capture or physical survey equipment. This gap represents the direct differentiator and primary thesis contribution opportunity. Implementation would require:

1. Physical testfield design on THWS campus (~0.5–2 hectares typical) with 8–12 GCPs + 4–6 QCPs measured via RTK GNSS or total station
2. UAV image acquisition via drone flight path crossing testfield
3. Parallel processing via: classical SfM/MVS pipeline and 3DGS pipeline
4. Metric validation: RMSE computation against GCP ground truth for both methods
5. Simultaneous evaluation of D1 through D5 dimensions on identical dataset

This approach would directly address the simulation-validation gap in GeoRefGS.

10.2 Secondary Research Gap: Cost-Speed-Accuracy Integration

While software accuracy benchmarks, processing time comparisons, and licensing cost analyses exist independently, no published study synthesizes all three dimensions simultaneously for UAV photogrammetry workflows. The DroneBundle industry analysis provides cost comparison, academic papers provide timing data, and ISPRS studies provide accuracy metrics—but integration across all three is absent. A thesis contribution could establish a comprehensive cost-speed-accuracy evaluation framework comparing classical and 3DGS approaches on identical hardware.

10.3 Tertiary Research Gap: ML Object Detection Performance

No systematic literature exists evaluating ML-based object detection performance on 3DGS-rendered views versus photogrammetric models. The 3DGS-QA perceptual quality dataset (225 degraded reconstructions, 15 object types) addresses subjective visual quality but not detection metrics (mAP). This represents a blank-slate evaluation dimension with potential for novel contribution.

10.4 Non-Gap (Addressed): Georeferencing Methodology

Georeferencing for 3DGS is no longer primarily a research gap. The convergence of GeoRefGS (academic), Pix4D commercial release, GPS-EXIF open-source pipeline, and glTF standards collectively address the integration question. The remaining gap is field validation, not methodology development.

10.5 Non-Gap (Confirmed): Skyfall-GS Applicability to UAV

Skyfall-GS is fundamentally satellite-only and requires substantial architectural modification for UAV application. This is not an open research question but a confirmed design constraint. Thesis framing should acknowledge this constraint in proposal phase.

11. Methodological Recommendations for Thesis Execution

11.1 Recommended Pipeline Stack

Classical Photogrammetry (Zero-Cost FOSS):

- Structure-from-Motion: COLMAP (open-source, command-line, reference standard)
- Multi-View Stereo: OpenMVS or OpenDroneMap (both open-source, UAV-optimized)
- Georeferencing: GPS-EXIF pipeline (GitHub: manudelu/georeferenced_gsplat) or Metashape Pro (THWS institutional license)
- Quality Assurance: CloudCompare (open-source, M3C2 analysis, Hausdorff distance, cloud-to-mesh distance)

3D Gaussian Splatting:

- Implementation: gsplat library (PyTorch+CUDA, 4× less memory than INRIA reference)
- NeRF baseline: Nerfstudio Splatfacto for direct comparison
- Commercial baseline: DJI Terra V5.0 (if Matrice 4E access available)
- Georeferencing: GeoRefGS framework (peer-reviewed, integrated GCP loss)
- Variant evaluation: DroneSplat (dynamic scene handling), SuGaR (mesh extraction), 2DGS (surface accuracy)

External Validation: GauU-Scene V2 dataset recommended for validation against established benchmarks with LiDAR ground truth.

11.2 Recommended Evaluation Metrics Matrix

Implement comprehensive simultaneous evaluation across D1–D5 dimensions:

D1 (Geometric Accuracy):

- RMSE_X, RMSE_Y, RMSE_Z separately against GCP ground truth (surveying standard)
- M3C2 distance via CloudCompare (statistical cloud-to-cloud comparison)
- Chamfer distance (symmetric distance metric for point cloud comparison)
- Cloud-to-Mesh distance (C2M) for object-scale evaluation

D2 (Visual Quality):

- PSNR (Peak Signal-to-Noise Ratio)
- SSIM (Structural Similarity Index)
- LPIPS (Learned Perceptual Image Patch Similarity via learned features)
- **Critical:** Always report all three simultaneously—GauU-Scene finding demonstrates potential anti-correlation between these metrics and geometric accuracy

D3 (Manual Object Identifiability):

- Structured human annotation protocol on rendered views at defined altitude increments
- Inter-rater reliability assessment (Cohen's kappa or Fleiss' kappa)
- Categories: clearly visible, partially obscured, barely discernible, not visible

D4 (ML Object Detection Performance):

- Apply standard detector (YOLOv8 recommended, well-established baseline) to rendered views from both methods at matched resolutions
- Report mean Average Precision (mAP) at IoU thresholds 0.5 and 0.75
- **Novel contribution:** This dimension has minimal existing literature

D5 (Practical Applicability):

- End-to-end wall-clock time (field capture through final deliverable)
- Peak VRAM consumption during training
- Final model size (MB)
- Software licensing cost (€/year amortized across projects)
- Hardware cost amortized per project

11.3 Thesis Timeline Considerations

Two execution pathways present different risk/contribution profiles:

Path A (Recommended for Bachelor Timeline):

- Reframe comparison to use drone-specific 3DGS (DroneSplat, standard INRIA implementation, or DJI Terra V5.0)
- Acknowledge Skyfall-GS as satellite-specific architectural reference
- Design physical campus testfield (0.5–2 hectares, 8–12 GCPs)
- Parallel processing via classical and 3DGS pipelines

- Comprehensive D1–D5 evaluation
- Timeline: Feasible within one academic year
- Risk level: Low-to-moderate

Path B (Maximum Contribution):

- Adapt Skyfall-GS architecture to oblique drone data
- Design physical testfield
- Explicit notation that UAV application constitutes novel application research
- Comprehensive evaluation including architectural modification justification
- Timeline: Requires careful scoping; potentially exceeds one-year thesis window
- Risk level: High; requires early supervisor feedback on feasibility

Path A recommended for initial proposal phase; Path B pursued only if early validation indicates technical feasibility.

12. Conclusion and Strategic Positioning

This literature review establishes a clear research positioning for the proposed thesis: the intersection of computer vision 3D reconstruction and geodetic measurement validation.

The current literature demonstrates that 3DGS and classical photogrammetry optimize for fundamentally different objectives. Classical SfM/MVS achieves superior metric geometric accuracy (1.43 mm RMSE in controlled conditions per ISPRS 2025), while 3DGS delivers substantially higher visual quality (PSNR, SSIM) and real-time rendering (100+ FPS). The GauU-Scene V2 finding of inverse correlation between rendering metrics and geometric accuracy contradicts the intuitive expectation that visual quality ensures geometric fidelity.

A genuine research gap exists: no published study has combined a physical GCP testfield measured by surveying instruments with 3DGS reconstruction and evaluated metric accuracy against surveying-grade ground truth. GeoRefGS, the closest published approach, validated exclusively in simulation. This gap is directly addressable through thesis work.

Skyfall-GS, cited in the research brief as the focal pipeline, is fundamentally architected for satellite imagery and has not been evaluated on UAV drone footage. This constraint must be resolved during proposal phase through scope clarification.

The thesis can make three distinct scientific contributions:

1. **Primary contribution:** Physical GCP testfield validation of 3DGS georeferencing accuracy, addressing the simulation-validation gap in existing literature
2. **Secondary contribution:** Comprehensive cost-speed-accuracy benchmark integrating processing time, hardware cost, and metric accuracy across classical and neural methods on identical UAV datasets
3. **Tertiary contribution:** ML object detection performance evaluation on reconstructed scenes—a dimension with minimal existing literature

The recommended execution pathway employs a drone-specific 3DGS baseline (DroneSplat or standard INRIA implementation) with classical SfM/MVS comparison on a physical THWS campus testfield measured via RTK GNSS or total station. This approach balances scientific rigor with

practical feasibility within a one-year bachelor's thesis timeline.

The document's seven topical blocks—3DGS fundamentals, Skyfall-GS assessment, classical photogrammetry, comparative studies, georeferencing, test field design, and benchmark datasets—provide the integrated knowledge base required for thesis planning and proposal development.